Project: Worked examples
Title: Pipe pressure drop and pump duty
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

Friction and fitting losses through series pipe segments at a given flow, the static head, the system curve, the duty point against a pump curve entered as points, the affinity-law shift for a speed or impeller change, and the NPSH available at the pump suction.
Q = Φ/(ρ c_p ΔT) when a heat load is given; v = Q/(πD²/4); Re = ρvD/μ; f = 64/Re (Re < 2300) else Colebrook–White 1/√f = −2 log10(ε/3.7D + 2.51/(Re√f)) by fixed-point iteration (≤ 100 steps) with the Swamee–Jain explicit value as start and cross-check; h_f = f (L + L_eq)/D · v²/2g + K v²/2g; H_sys(Q) = Σdz + h_out + Σh_f(Q) with h_f ∝ Q² scaled from the design flow (closed circuit: Σdz = 0, the rise in the flow is recovered in the return); the water held in the pipes V_pipe = Σ π/4·D²·L; duty point where H_pump(Q) = H_sys(Q) (bisection); affinity: Q ∝ n D³, H ∝ n² D²; NPSH_a = p_atm/ρg + z_s − h_f,suction − p_v/ρg.
Ref: Crane Technical Paper 410 — Flow of fluids through valves, fittings and pipe
Ref: Çengel & Cimbala, Fluid Mechanics Ch. 8 — Internal flow: friction factor, minor losses, pump duty
Ref: CIBSE Guide C Section 4 — Flow of fluids in pipes and ducts (principles)
Ref: Kell, J. Chem. Eng. Data 20(1), 1975 — Density of liquid water

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
fluid	Fluid	water		spec: SPEC-M-04 §3: chilled water, no glycol	any
T	Fluid temperature (mean; link B6 t_m for a heating loop)	6	degC	spec: SPEC-M-04 §3: 6 °C flow	0 … 100 degC
Q	Design flow rate (blank when a heat load gives it)	30	m^3/h	spec: Schedule CHW-01 design flow	0.01 … 100000 m^3/h
segments	Pipe segments in series	3 row(s)		drawing: DRG-M-110 rev C	any
segments[1]		name = plantroom suction; L = 8 m; D = 100 mm; eps = 0.045 mm; K = 3; L_eq = 0 m; dz = 0 m; side = suction			

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
segments[2]		name = riser DN80; L = 60 m; D = 80 mm; eps = 0.045 mm; K = 8; L_eq = 0 m; dz = 10 m; side = discharge			
segments[3]		name = index run DN65; L = 45 m; D = 65 mm; eps = 0.045 mm; K = 12; L_eq = 6 m; dz = 2 m; side = discharge			
circuit	Circuit (a closed flow-and-return loop has no static lift: the rises entered are ignored)	open (the pump lifts the water by the rises entered)		default	any
h_out	Pressure head required at the outlet	3	m	spec: Terminal unit: 30 kPa ≈ 3 m	0 … 1000 m
v_max	Velocity limit (warning)	2.5	m/s	spec: CIBSE Guide C practice limit, assumed	0.5 … 10 m/s
Q_pump	Pump curve: flow points	[0, 20, 40, 60]	m^3/h	spec: Pump datasheet P-01 rev 2, curve points	any
H_pump	Pump curve: head points	[40, 36, 24, 4]	m	spec: Pump datasheet P-01 rev 2, curve points	any
n_ratio	Speed ratio n_2/n_1 (affinity)	0.9		assumption: Inverter turn-down to 90 % speed.	0.2 … 2
D_ratio	Impeller diameter ratio D_2/D_1 (affinity)	1		assumption: Impeller unchanged.	0.7 … 1.1

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
eta_pump	Pump efficiency at duty (for shaft power)	0.68		spec: Pump datasheet P-01 rev 2	0.2 … 0.95
p_atm	Atmospheric pressure (NPSH)	101.325	kPa	assumption: Sea level.	50 … 110 kPa
z_s	Static head at the pump suction (+ flooded, − lift)	2	m	drawing: DRG-M-110 rev C	-10 … 100 m
NPSH_r	NPSH required by the pump (optional)	3	m	spec: Pump datasheet P-01 rev 2	0 … 50 m

3. Assumptions

A1. Steady, incompressible, fully developed flow; all segments carry the full design flow in series. (Darcy–Weisbach.)
A2. Fitting losses scale with velocity head (K method) or add pipe length (equivalent-length method); both may be used on one segment. (Crane TP-410 practice.)
A3. The system curve scales friction losses with Q^2 from the design flow (turbulent assumption); the static part is independent of flow. ($H_{sys} = H_{static} + k\,Q^2$.)
A4. Water properties are from published correlations at atmospheric pressure (0–100 °C); any other fluid is entered with its own density and viscosity and their provenance. (Kell 1975; Vogel fit; Hyland & Wexler 1983.)
A5. n_ratio (Speed ratio n_2/n_1 (affinity)) is taken as 0.9. (Inverter turn-down to 90 % speed.)
A6. D_ratio (Impeller diameter ratio D_2/D_1 (affinity)) is taken as 1. (Impeller unchanged.)
A7. p_atm (Atmospheric pressure (NPSH)) is taken as 101.325 kPa. (Sea level.)

4. Working

[3] rho = Kell (1975) density of water at T — Kell 1975
= T = 6 °C
= 999.94 kg/m³
Density
[4] mu = Vogel fit □ = 0.02939 exp(507.88/(T_K - 149.3)) mPa·s
= T_K = 279.15 K
= 1.4684 mPa·s
Dynamic viscosity
[5] p_v = Hyland–Wexler saturation pressure
= T = 6 °C
= 0.93525 kPa
Vapour pressure (NPSH)
[6] T_line = T (no T_line entered)
= 6 °C
= 6 degC
Design temperature of the line to insulate: the fluid temperature T, since no T_line was entered
[10] Q_design = Q
= 30 m³/h
= 30 m³/h
Design flow (as entered)
[11] v_s1 = plantroom suction: Q / (π D²/4)
= 30 m³/h / (π × 100² mm² / 4)
= 1.061 m/s
Velocity, plantroom suction

[12] Re_s1 = plantroom suction: ρ v D / μ
= 999.94 × 1.061 × 0.1 / 0.0014684
= 72253
Reynolds number, plantroom suction
[13] f_s1 = plantroom suction: Darcy friction factor
= Colebrook–White, 10 iterations (Swamee–Jain start 0.021152)
= 0.021093
Friction factor, plantroom suction
[14] f_sj_s1 = plantroom suction: Swamee–Jain check
= 0.25 / log10(ε/3.7D + 5.74/Re^0.9)²
= 0.021152
Explicit cross-check, plantroom suction
[15] h_f_s1 = plantroom suction: f (L + L_eq)/D · v²/2g + K v²/2g
= 0.021093 × (8 + 0) / 0.1 × 0.057399 + 3 × 0.057399
= 0.26906 m
Head loss, plantroom suction
[16] v_s2 = riser DN80: Q / (π D²/4)
= 30 m³/h / (π × 80² mm² / 4)
= 1.6579 m/s
Velocity, riser DN80
[17] Re_s2 = riser DN80: ρ v D / μ
= 999.94 × 1.6579 × 0.08 / 0.0014684
= 90316
Reynolds number, riser DN80
[18] f_s2 = riser DN80: Darcy friction factor
= Colebrook–White, 10 iterations (Swamee–Jain start 0.020946)
= 0.020849
Friction factor, riser DN80
[19] f_sj_s2 = riser DN80: Swamee–Jain check
= 0.25 / log10(ε/3.7D + 5.74/Re^0.9)²
= 0.020946
Explicit cross-check, riser DN80
[20] h_f_s2 = riser DN80: f (L + L_eq)/D · v²/2g + K v²/2g
= 0.020849 × (60 + 0) / 0.08 × 0.14014 + 8 × 0.14014
= 3.3123 m
Head loss, riser DN80
[21] v_s3 = index run DN65: Q / (π D²/4)
= 30 m³/h / (π × 65² mm² / 4)
= 2.5113 m/s
Velocity, index run DN65
[22] Re_s3 = index run DN65: ρ v D / μ
= 999.94 × 2.5113 × 0.065 / 0.0014684
= 111160
Reynolds number, index run DN65
[23] f_s3 = index run DN65: Darcy friction factor
= Colebrook–White, 9 iterations (Swamee–Jain start 0.020964)
= 0.020834
Friction factor, index run DN65
[24] f_sj_s3 = index run DN65: Swamee–Jain check
= 0.25 / log10(ε/3.7D + 5.74/Re^0.9)²
= 0.020964
Explicit cross-check, index run DN65

[25] h_f_s3 = index run DN65: f (L + L_eq)/D · v²/2g + K v²/2g
= 0.020834 × (45 + 6) / 0.065 × 0.32155 + 12 × 0.32155
= 9.1151 m
Head loss, index run DN65
[26] h_f_total = Σ h_f
= 3 segment(s)
= 12.696 m
Total friction and fitting loss at the design flow
[28] V_pipe = Σ π/4·D²·L
= 3 segment(s)
= 513.75 L
Water held in the pipe segments
[29] H_static = Σ dz + h_out
= 12 + 3
= 15 m
Static head
[30] H_sys = H_static + h_f_total
= 15 m + 12.696 m
= 27.696 m
System head at the design flow
[31] dp_total = rho * 9.80665 m/s^2 * h_f_total
= 999.94 kg/m³ × 9.80665 m/s^2 × 12.696 m
= 124.5 kPa
Friction pressure drop at the design flow
[32] dp_sys = rho * 9.80665 m/s^2 * H_sys
= 999.94 kg/m³ × 9.80665 m/s^2 × 27.696 m
= 271.59 kPa
System pressure at the design flow
[33] P_hyd = rho * 9.80665 m/s^2 * Q_design * H_sys
= 999.94 kg/m³ × 9.80665 m/s^2 × 30 m³/h × 27.696 m
= 2.2633 kW
Hydraulic power at the design point
[39] Q_duty = duty flow (as entered)
= H_pump(Q) = H_sys(Q), bisection
= 31.569 m³/h
Duty flow as entered
[40] H_duty = duty head (as entered)
= H_sys(31.569 m³/h)
= 29.059 m
Duty head as entered
[41] P_shaft = Σ g Q H / η
= 999.94 × 9.8066 × 0.008769 × 29.059 / 0.68
= 3.6747 kW
Shaft power at duty as entered
[42] affinity = $Q_2 = Q_1 (n_2/n_1)(D_2/D_1)^3$, $H_2 = H_1 (n_2/n_1)^2(D_2/D_1)^2$
= $n_2/n_1 = 0.9$, $D_2/D_1 = 1$
= "curve scaled"
Affinity laws applied to the pump curve
[43] Q_duty2 = duty flow (after the affinity shift)
= H_pump(Q) = H_sys(Q), bisection
= 26.238 m³/h
Duty flow after the affinity shift

[44] H_duty2 = duty head (after the affinity shift)
= H_sys(26.238 m³/h)
= 24.712 m
Duty head after the affinity shift
[45] P_shaft2 = □ g Q H / □
= 999.94 × 9.8066 × 0.0072883 × 24.712 / 0.68
= 2.5972 kW
Shaft power at duty after the affinity shift
[48] h_f_suction = □ h_f on the suction side
= 1 suction segment(s)
= 0.26906 m
Suction-side friction loss
[49] NPSH_a = p_atm / (rho * 9.80665 m/s^2) + z_s - h_f_suction - p_v / (rho * 9.80665 m/s^2)
= 101.32 kPa / (999.94 kg/m³ × 9.80665 m/s^2) + 2 m − 0.26906 m − 0.93525 kPa / (999.94 kg/m³ × 9.80665 m/s^2)
= 11.968 m
NPSH available
[51] u_npsh = NPSH_r / NPSH_a
= 3 m / 11.968 m
= 0.25066
NPSH utilisation (required / available)
[52] v_peak = highest segment velocity
= 2.5113 m/s
= 2.5113 m/s
Highest velocity
[53] u_v = v_peak / v_max
= 2.5113 / 2.5
= 1.0045
Velocity utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
T_line	6 degC	Line design temperature
v_s1	1.061 m/s	Velocity, plantroom suction
Re_s1	72253	Re, plantroom suction
f_s1	0.021093	f, plantroom suction
f_sj_s1	0.021152	f (Swamee–Jain), plantroom suction
h_f_s1	0.26906 m	h_f, plantroom suction
v_s2	1.6579 m/s	Velocity, riser DN80
Re_s2	90316	Re, riser DN80
f_s2	0.020849	f, riser DN80
f_sj_s2	0.020946	f (Swamee–Jain), riser DN80
h_f_s2	3.3123 m	h_f, riser DN80
v_s3	2.5113 m/s	Velocity, index run DN65
Re_s3	111160	Re, index run DN65
f_s3	0.020834	f, index run DN65

SYMBOL	VALUE	DESCRIPTION
f_sj_s3	0.020964	f (Swamee–Jain), index run DN65
h_f_s3	9.1151 m	h_f, index run DN65
h_f_total	12.696 m	Σ h_f
H_static	15 m	Static head
V_pipe	513.75 L	Water held in the pipes
H_sys	27.696 m	System head at Q
dp_total	124.5 kPa	Friction Δp
dp_sys	271.59 kPa	System Δp
Q_design	30 m³/h	Design flow
P_hyd	2.2633 kW	Hydraulic power
Q_duty	31.569 m³/h	Duty flow as entered
H_duty	29.059 m	Duty head as entered
P_shaft	3.6747 kW	Shaft power as entered
Q_duty2	26.238 m³/h	Duty flow after the affinity shift
H_duty2	24.712 m	Duty head after the affinity shift
P_shaft2	2.5972 kW	Shaft power after the affinity shift
NPSH_a	11.968 m	NPSH available
v_peak	2.5113 m/s	Highest velocity

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Velocity ≤ limit	u_v = 1.0045	u_v ≤ 1	1	
PASS	NPSH_r ≤ NPSH_a	u_npsh = 0.25066	u_npsh ≤ 1	0.251	
PASS	Duty point exists (as entered)	31.569 m³/h	29.059 m	0	
PASS	Duty point exists (after the affinity shift)	26.238 m³/h	24.712 m	0	

7. Applicability

WARNING — input.circuit: circuit (Circuit (a closed flow-and-return loop has no static lift: the rises entered are ignored)) was not entered; the template default 'open (the pump lifts the water by the rises entered)' is used.
WARNING — velocity: segments: index run DN65 runs at 2.5113 m/s, above the 2.5 m/s limit.

8. Verification

Template bs.pipe-pump@1.5.0 Content hash fa61fbf098961063 Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0